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The Late Middle Pleistocene Hominin Fossil Record of Eastern Asia: Synthesis and Review

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ABSTRACT Traditionally, Middle Pleistocene hominin fossils that cannot be allocated to *Homo erectus* sensu lato or modern *H. sapiens* have been assigned to different specific taxa. For example, in eastern Asia, these hominin fossils have been classified as archaic, early, or premodern *H. sapiens*. An increasing number of Middle Pleistocene hominin fossils are currently being assigned to *H. heidelbergensis*. This is particularly the case for the African and European Middle Pleistocene hominin fossil record. There have been suggestions that perhaps the eastern Asian late Middle Pleistocene hominins can also be allocated to the *H. heidelbergensis* hypodigm. In this article, I review the current state of the late Middle Pleistocene hominin fossil record from eastern Asia and examine the various arguments for assigning these hominins to the different specific taxa. The two primary conclusions drawn from this review are as follows: 1) little evidence currently exists in the eastern Asian Middle Pleistocene hominin fossil record to support their assignment to *H. heidelbergensis*; and 2) rather than add to the growing list of hominin fossil taxa by using taxonomic names like *H. daliensis* for northeast Asian fossils and *H. mabaensis* for Southeast Asian fossils, it is better to err on the side of caution and continue to use the term archaic *H. sapiens* to represent all of these hominin fossils. What should be evident from this review is the need for an increase in the quality and quantity of the eastern Asian hominin fossil data set. Fortunately, with the increasing number of large-scale multidisciplinary paleoanthropological field and laboratory research projects in eastern Asia, the record is quickly becoming better understood. Yrbk Phys Anthropol 53:75–93, 2010. ©2010 Wiley-Liss, Inc.

"[T]he 'variation' regarded as permissible within a single hominid species became expanded beyond all reason. Any signal of systematic diversity was lost. Perhaps the most pernicious, and certainly the most pervasive of all the various devices invented to sustain this new orthodoxy, was 'archaic *Homo sapiens*.'" (Tattersall and Schwartz, 2008, p 50).

"The obviously great diversity in using the name *Homo heidelbergensis* reveals that it is hardly a well-defined taxon" (Brauer, 2008, p 32), thus leading Brauer (2008, p 35) to conclude that "the use of archaic *Homo sapiens* still appears adequate and plausible."

The nature of hominin biological evolution during the Middle Pleistocene is one of the most debated topics in paleoanthropology today. Debates range from lumpers questioning the validity of *H. erectus* as a separate species from *H. sapiens* (e.g., Wolpoff et al., 1984, 1994; Wolpoff and Caspari, 1997; Wolpoff, 1999) to splitters arguing as many as seven different hominin taxa (*H. erectus*, *H. antecessor*, *H. heidelbergensis*, *H. rhodesiensis*, *H. helmei*, *H. neandertalensis*, and *H. sapiens*) roamed the Old World during the Middle Pleistocene, some that were apparently coeval (for discussion see syntheses published in just the last decade by Stringer, 2002; Conroy, 2005; Brauer, 2008; Rightmire, 2008; Wood and Lonergan, 2008; Cartmill and Smith, 2009; Klein, 2009).[1,2] Since the 1960s paleoanthropologists have allocated hominins from the Middle Pleistocene that predate Neandertals and modern humans, but display more derived features than *H. erectus*, into the category "archaic" *H. sapiens*. Cartmill and Smith (2009, p 293) note that these archaic *H. sapiens* were "admitted to membership in our species because of their almost modern-sized brains, but set off as 'archaic' because of their primitive-looking cranial morphology." Archaic *H. sapiens* is sometimes divided into early and late groups, based primarily on chronological grounds.

Nevertheless, Rightmire (1998, p 218) observed more than a decade ago that "[i]t is apparent that the traditional approach of lumping diverse humans together as 'archaic' *Homo sapiens* will no longer work. The picture is highly complex, and several taxa probably are needed to accommodate the fossils." *H. heidelbergensis* is the taxonomic name becoming increasingly more common when referring to these "transitional" Middle Pleistocene hominins (Rightmire, 2008; Tattersall and Schwartz, 2008). Often included in the *H. heidelbergensis* hypodigm are fossils from Africa (e.g., Bodo, Broken Hill, Ndutu, and Elandsfontein) and Europe (e.g., Mauer, Arago, Petralona, Boxgrove, Sima de los Huesos [Atapuerca]) and possibly Southwest Asia (e.g., Zuttiyeh) (Conroy, 2005; Rightmire, 2008; Cartmill and Smith,

[1]In paleontology, "lumpers" are scholars that emphasize similarities between fossils, "splitters" emphasize differences (for discussion see Kimbel, 1991; Conroy, 2002; Wood and Lonergan, 2008 among others).

[2]Some might add *Homo georgicus* to this list, but currently only the Early Pleistocene Dmanisi hominin fossils have been allocated to this taxon, thus, falling outside the scope of this article.

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2009; Klein, 2009). The holotype of *H. heidelbergensis* is the Mauer mandible, which was discovered in Germany in 1907 and originally described by Otto Schoetensack (1908). Recent detailed morphometric comparative analysis of the Mauer mandible found support for *H. heidelbergensis* as a valid stand-alone species (Mounier et al., 2009). Although the *H. heidelbergensis* holotype is a mandible with some apparent pathologies (Czarnetzki et al., 2003), similarities between it and other Middle Pleistocene mandibles (e.g., Arago 2, 13, and Tighenif 3) have been identified (Mounier et al., 2009). In turn, similarities between the crania from Arago, Petralona, and Broken Hill suggest to some (e.g., Rightmire, 2008) that all of these fossils belong to the *H. heidelbergensis* hypodigm. For example, Rightmire (1998, p 221) observes that the Petralona and Broken Hill crania have similarities in cranial "height, breadth, and massive construction of the upper face and cheek, several measures of projection in the facial midline, configuration of the thickened brows, and many aspects of vault shape." The case for *H. heidelbergensis* as a distinct taxon that represents many of the Middle Pleistocene hominin fossils from at least Africa and Europe is well documented (e.g., Rightmire, 1998, 2008; Stringer, 2002).

A number of cranial characteristics are recognized to be distinctive of *H. heidelbergensis* (as reviewed recently by Rightmire, 2008). For example, its cranial capacity (~1,200 cm^3) is intermediate between *H. erectus* (~1,000 cm^3) and *H. sapiens* (~1,350 cm^3). When brain size is scaled to body mass, it is also intermediate between *H. erectus* and *H. sapiens* (Rightmire, 2004). With the increase in *H. heidelbergensis* cranial capacity, there is a reduction in the degree of postorbital constriction, eventually leading to modern *H. sapiens*-like morphology. *H. heidelbergensis* also has a more rounded and less-angled occipital and a greater degree of cranial base flexion than *H. erectus* (but less than modern *H. sapiens*). The three primary characters often noted about the *H. heidelbergensis* mandibles, particularly the Mauer holotype are: 1) its robustness; 2) substantial anteroposterior mandibular ramus length; and 3) absence of a mental eminence, i.e., chin (Mounier et al., 2009). However, variation in these characters exists within the *H. heidelbergensis* hypodigm. For instance, the Arago 13 hemimandible displays elements of a mental eminence (Rightmire, 2008; see detailed review of the hominin fossil evidence for the "chin" by Schwartz and Tattersall, 2000). It should be noted, however, that none of these character traits are viewed as true *H. heidelbergensis* autapomorphies, but rather as gradistic[3] changes between *H. erectus* and *H. sapiens* (Rightmire, 2008).

The evolutionary position of *H. heidelbergensis* remains unclear (Groves and Lahr, 1994; Lahr and Foley, 1998; Rightmire, 1998, 2004, 2008). The evidence from Europe and Africa currently favors the scenario where *H. erectus* is ancestral to *H. heidelbergensis*, which in turn gave rise to *H. neandertalensis* in Europe and modern *H. sapiens* in Africa (Stringer, 2002; Rightmire, 2008). A second scenario argues that *H. erectus* is ancestral to *H. antecessor*, which in turn gave rise to *H. heidelbergensis* in Europe and *H. rhodesiensis* in Africa (Bermudez de Castro et al., 1997; Arsuaga et al., 1999). Yet a third scenario excludes *H. heidelbergensis* and favors the view that *H. rhodesiensis*/*H. helmei* in Africa gave rise to *H. neandertalensis* in Europe and modern *H. sapiens* in Africa (for review see Lahr and Foley, 1998; Stringer, 2002; Rightmire, 2008: his Fig. 5).

What is not clear from the above discussion is the role the eastern Asian Middle Pleistocene hominin fossil record plays. More than one decade ago, Rightmire (1998, p 225) remarked that "[w]hether the [Chinese] skeletons should be lumped with *Homo heidelbergensis* is one issue; how they are related to recent Asian populations is another. Both are fraught with controversy." Furthermore, Rightmire (2004, p 109) more recently noted that "[i]f *Homo heidelbergensis* is considered (sensu lato) to include populations from Africa as well as Europe, there is still some doubt in respect to the Far East." Because of the paucity of recent syntheses of the eastern Asian Middle Pleistocene hominin fossil record, I suggest that Rightmire's observations are still relevant and worthy of more detailed investigation. In particular, the current state of the eastern Asian late Middle Pleistocene hominin fossil record needs to be assessed to better understand the role the hominin fossil record from this spatiotemporal facie plays in addressing the question of what name these hominins should be referred to.

Most paleoanthropologists familiar with the eastern Asian record (e.g., Wolpoff et al., 1984; Pope, 1991, 1992; Wu and Poirier, 1995; Etler, 1996, 2004) often use the terms "archaic," "early," or "premodern" to refer to hominin fossils that are not quite modern in morphology, but yet would not be considered "classic" *H. erectus* sensu lato. However, it has been proposed that these eastern Asian archaic humans can be readily assigned to *H. heidelbergensis* (Groves and Lahr, 1994; Lahr, 1996; Stringer, 2002). Nevertheless, Pope (1992, p 251) suggests that it "is important to continue to employ separate terminologies for these widely separated geographic groups because there are substantial morphological differences between the Archaic Euro-African and Neanderthals on the one hand and the Premodern Chinese hominids on the other."

The focus of this article was to review the current state of the eastern Asian late Middle Pleistocene hominin fossil record, focusing on those fossils that are in taxonomic limbo somewhere between *H. erectus* and modern *H. sapiens*. This article provides a synthesis of the available data, particularly by building on the seminal works of Pope (1992), Wu and Poirier (1995), and Etler (1996). I evaluate the arguments that 1) the late Middle Pleistocene hominin fossils from eastern Asia should be included within the *H. heidelbergensis* hypodigm, 2) whether a different taxonomic name might be more apropos, or 3) whether the archaic *H. sapiens* classification should continue to be used. Although I mention modern human origins when applicable, I do not review the arguments for regional continuity between *H. erectus* and modern *H. sapiens* (with archaic *H. sapiens* being the transitional group) or the replacement of the former by the latter in eastern Asia. These human origins models are well documented elsewhere (e.g., Weidenreich, 1943; Wolpoff et al., 1984, 2001; Cann et al., 1987; Stringer and Andrews, 1988; Pope, 1992; Frayer et al., 1993; Etler, 1996, 2004; Lahr, 1996; Hawks et al., 2000; Stringer, 2002; Templeton, 2002, 2005; Trinkaus, 2005; Curnoe, 2007).

[3]In evolutionary biology, grades are defined as groups of organisms that share characters, regardless of how closely related they are. Clades are groups of organisms that share characters, but a common ancestor as well (see Simpson, 1961; Wolpoff et al., 1984; Futuyma, 1998 among others for discussion).

MIDDLE PLEISTOCENE HOMININS IN EASTERN ASIA

Broadly speaking, eastern Asia can be topographically divided into two regions by the Himalayan and Qinling mountain ranges, which effectively run through western and central China. The major uplift of the Himalayas occurred sometime during the Neogene (Fort, 1996; Dennell, 2009), which served as a formidable barrier throughout the Quaternary. The lower-lying Qinling mountain range, which is east of the Himalayas, would also have formed a barrier. To the east of the Qinling mountains, the region of eastern China is low-lying, with much of the region less than 1,000 m above sea level (Norton and Jin, 2009; Norton et al., 2010a,b). Because of this unique topography, eastern Asia can be divided into two primary regions: 1) Siberia, Mongolia, northern China, Korea, and Japan; and 2) southern China and mainland and insular Southeast (SE) Asia. In this article, I refer to the former area as Northeast Asia and the latter region as SE Asia. The primary distinction between the two areas is that the northern part falls within the Palearctic biogeographic region and the southern part represents the Oriental biogeographic region. Biogeographically, insular SE Asia is separated from Australasia by Wallace's Line. I have argued elsewhere (Norton et al., 2010a,b) that the low-lying region of eastern China would have served as a corridor where cold-adapted faunas would have migrated southward during stadials and warm-adapted taxa northward during interstadials. Indeed, presence of normally Oriental region-restricted nonhuman primates in Middle Pleistocene localities in northern China and Korea seem to support this argument (Jablonski et al., 2000; Norton, 2000; Norton et al., 2010b).

Because of the presence of fossil hominins in two distinct biogeographic zones (Palearctic and Oriental), the question that might arise is should we expect to find regional variation in hominin morphology as studies of eastern Asian human fossil materials have shown (e.g., Turner, 1990; Wu et al., 2007; Pietrusewsky, 2010)? Namely, might we distinguish the Middle Pleistocene hominin fossils from SE Asia (including southern China) and Northeast Asia, with morphologically transitional specimens present along the fluctuating Palearctic/Oriental boundary? Morphometric analysis of the Middle Pleistocene Tangshan, Huludong *H. erectus* crania (Liu et al., 2005), which is located in Nanjing, Jiangsu Province (central-east China), right along the Palearctic/Oriental boundary suggests this may be the case (see also Anton, 2002). The Liu et al. (2005) study found that the Tangshan *H. erectus* fossils share many features of the coeval Zhoukoudian (ZKD) Locality 1 hominins, but also had a few characters similar to the Indonesian *H. erectus*. A similar discussion has been presented for the nearby Hexian *H. erectus* cranium that seems to display character traits that could align it with the late *H. erectus* Ngandong hominins (Huang et al., 1982), but more likely ZKD *H. erectus* (Pope, 1992; Wu et al., 2006). Because of this biogeographic distinction, I divide this review of the hominin fossil record into Northeast and SE Asia (Fig. 1 and Table 1).

Important Northeast Asian hominin fossil localities

In addition to the well-known *H. erectus* fossils from ZKD Locality 1, a number of important Middle Pleistocene archaic *H. sapiens* sites exist in Northeast Asia, primarily from China (Pope, 1992; Etler and Li, 1994; Wu and Poirier, 1995; Etler, 1996). Middle and Late Pleistocene hominin fossils have also been reported from Mongolia and Korea (Norton, 2000; Coppens et al., 2008). The best-described NE Asian archaic *H. sapiens* are from Dali, Jinniushan, and Xujiayao, which are located in present-day China.

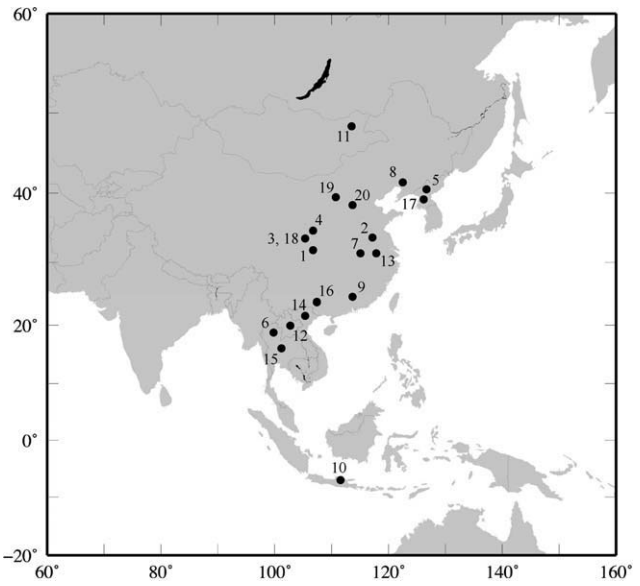


Fig. 1. Locations of paleoanthropological sites mentioned in the text. 1. Changyang, 2. Chaoxian, 3. Chenjiawo, 4. Dali, 5. Danyang Kunangul, 6. Dokchon Soongnisan, 7. Had Pu Dai, 8. Hexian, 9. Jinniushan, 10. Maba, 11. Ngandong, 12. Salkhit, 13. Tam Hang, 14. Tangshan, 15. Tham Kuyen, Ma U'Oi, 16. Thum Wiman Nakin, 17. Tongzi, 18. Yokpo Daehyundong, Ryonggok, Mandalli, 19. Yunxian, 20. Xujiayao, and 21. Zhoukoudian.

Dali is an open-air locality situated on the third terrace of the Lohe River located in Shaanxi Province, which is in the Chinese Loess Plateau (Wu and Poirier, 1995). Dali is represented by one almost complete skull that was discovered in 1978. The skull has been described as a robust male approximately 30 years of age. The cranium was found in association with a variety of typical Palearctic taxa and small stone flake tools. Interestingly, warm, humid-adapted faunal elements (e.g., *Bubalus*) were present, in association with taxa usually found in cooler, more forested environments (e.g., *Palaeoloxodon*). The only reported uranium-series date for Dali is 209 ± 23 ka (Chen et al., 1984) and a more recent paleomagnetic dating study brackets the hominin fossil between 300 and 260 ka (Yin et al., 2002). Pope (1992), noting the rolled and abraded condition of the paleontological and archaeological materials, suggests that the collection accumulated as the result of fluvial activity.

Wu (1981) and others (e.g., Pope, 1992) observed that the Dali skull has a number of traits that can be found in western archaic *H. sapiens* (e.g., Steinheim, Arago, and Jebel Irhoud), eastern *H. erectus*, and modern *H. sapiens*. For example, the thickness of the frontal and temporal squamae and parietal tuberosity fall within the range of ZKD Locality 1 *H. erectus*. In addition, the orbits are quadrangular, a characteristic common in *H. erectus* (Wu and Poirier, 1995). However, the degree of postorbital constriction and thickness and concavity of

TABLE 1. Eastern Asian paleoanthropological localities discussed in text (see Fig. 1 for locations)

Locality	Country	Open-air/ cave	Age	Dating technique	Hominin fossils	Cultural assignment	Archaeological remains	Faunal remains	Reference(s)
Salkhit	Mongolia	Open-air	Late Pleistocene	Biostratigraphy	Assorted cranial fragments	None reported	None reported	Palearctic	Coppens et al. (2008)
Jinniushan	China	Cave	~280–260 ka	U-series, ESR	Assorted cranial and postcranial fragments representing single individual	Early Paleolithic	~200 lithic artifacts	Palearctic	Lu (1989, 2003), Rosenberg et al. (2006)
Xujiayao	China	Open-air	~500–16 ka	Magnetostratigraphy, U-series, ^{14}C	Assorted cranial fragments	Early Paleolithic	Small scrapers, points, and burins	Palearctic	Wu and Poirier (1995), Norton and Gao (2008a)
Dali	China	Open-air	209 ka	U-series	Almost complete cranium	Early Paleolithic	Small scrapers, points, and burins	Palearctic	Chen et al. (1984), Wu and Poirier (1995)
Yokpo Daehyundong	Korea	Cave	Middle/Late Pleistocene	Biostratigraphy	Cranial fragments	None reported	N/A	Palearctic	Kim et al. (1985), Norton (2000)
Ryonggok Cave	Korea	Cave	500–400 ka/ 48–46 ka	TL, U-series, biostratigraphy		Early Paleolithic	Core and flake tools	Palearctic	Jun et al. (1986), Norton (2000)
				Layer 9	Cranium no. 7, mandibles nos. 1, 2, and 6				
				Layer 10	Cranium no. 3, mandible nos 4, 6, and femur				
				Layer 11	Various postcranial fragments				
				Layer 12	Maxilla no. 8				
Dokchon, Soongnisan	Korea	Cave	Middle/Late Pleistocene	Biostratigraphy	Two molars, scapula, mandible	None reported	N/A	Palearctic	Archaeology Research Laboratory, (1978), Norton (2000)
Maba	China	Cave	~237 ka	Biostratigraphy, U-series	Skull, mandible, assorted loose teeth	None reported	N/A	Oriental	Wu and Poirier (1995), Gao et al. (2007)
Chaoxian	China	Cave	~360–310 ka	Biostratigraphy, U-series	Cranial fragments, partial maxilla, loose teeth	None reported	N/A	Oriental	Wu and Poirier, (1995), Shen et al. (2010)
Ma U'Oi	Vietnam	Cave	~193–49 ka	Biostratigraphy, U-series	Loose teeth	None reported	N/A	Oriental	Demeter et al. (2004, 2005)

the tympanic plate lie between *H. erectus* and modern *H. sapiens*. Additionally, the cranial capacity of 1,120 cm^3 falls between *H. erectus* and *H. sapiens*. A number of measurements and indices derived from the Dali cranium also fall within the range of archaic *H. sapiens* and are intermediate between *H. erectus* and modern *H. sapiens*. These include maximum length, maximum width, and transverse arc and curvature, which are more characteristic of archaic *H. sapiens* and outside the range of *H. erectus* (see Wu and Poirier, 1995: their Table 3.1).

In 1984, a partial hominin skeleton was discovered during excavations at Jinniushan, a karst fissure/collapsed cave site located in Liaoning Province, northeastern China (Lu, 1989, 2003; Wu and Poirier, 1995; Rosenberg et al., 2006). The Jinniushan skeletal collection comprises a more or less complete cranium (fragmentary, but undistorted) and ~50 assorted partial and complete postcranial elements. Almost all of the fossils were excavated from a small 1.6 m^2 area within the cave, suggesting that all of the bones derive from one individual (Lu, 2003). The hominin fossils were found in association with a mixture of Palearctic (e.g., *Dicerorhinus*) and Oriental (e.g., *Macaca*) taxa as well as about 200 lithic artifacts produced on locally available quartz and siliceous limestone. An array of U-series and electron spin resonance dates suggests the Jinniushan hominin may date to ~280–260 ka (Lu, 2003), though some (e.g., Pope, 1992) have questioned the relationship between the hominin fossils and the samples used for dating.

The Jinniushan skeleton was originally interpreted to be an adult male (Wu, 1988a; Lu, 1989). However, Jinniushan has also been determined to be a female, based on the morphology of the pubis which displays a subpubic concavity and the "medial aspect of the ischiopubic ramus [which] is ridged rather than flat" (Rosenberg et al., 2006, p 3552). These traits are considered to be the two primary morphologies of the pubis that distinguish sex (Phenice, 1969). The Jinniushan cranium is considered to be larger, but more gracile than the Dali skull. For example, the cranial walls and brow ridges are less robust than *H. erectus* or Dali. The Jinniushan estimated cranial capacity of ~1,300 cm^3 is within the range of modern humans.

Rosenberg et al. (2006) conducted the most recent morphometric analysis of the Jinniushan materials and drew a number of interesting conclusions. In particular, they used a series of regression equations to estimate height, weight, and body proportions and determined that the Jinniushan female was ~168 cm tall and ~77 kg in weight. Given the fact that Jinniushan is located in a high-latitude region, and given what we know of ecogeographic clines in body size and structure (sensu Bergmann's and Allen's Rules), the results of the Rosenberg et al.'s (2006) study conform to what might be expected of a female hominin living in this type of environment. To date, the Jinniushan hominin is the largest female that predates Holocene modern humans and falls within range of body size reconstructions from other Middle Pleistocene hominins (e.g., Boxgrove, Atapuerca) (Rosenberg et al., 2006).

Xujiayao is an open-air site located in the western part of the Nihewan Basin in Shanxi Province, northern China (Jia et al., 1979; Wu and Poirier, 1995; Norton and Gao, 2008a). The site was discovered during field surveys conducted by the Institute of Vertebrate Paleontology and Paleoanthropology, Chinese Academy of Sciences in 1974 and subsequently excavated in 1976, 1977, and 1979 (Jia et al., 1979; Norton and Gao, 2008a). The hominin fossils were found in association with a diversity of Early Paleolithic artifacts (Gao and Norton, 2002) and a faunal collection dominated by *Equus*, with a smaller number of artiodactyla (e.g., *Spiroceros*, *Gazella*). Taphonomic analysis of the associated faunal assemblage suggested abundant evidence of efficient hunting and butchering of horse carcasses (Norton and Gao, 2008a). The chronometric age of the Xujiayao deposit is unclear, with ages ranging from ~16 ka (^{14}C) to ~500 ka (magnetostratigraphy) (Norton and Gao, 2008a). The majority of the chronometric studies suggest an early Late Pleistocene age, though there is no clear consensus (Norton and Gao, 2008a; Nagatomo et al., 2009).

Xujiayao is best known for the presence of an array of hominin fossils, including twelve parietals, two occipitals, and one temporal fragment that have been assigned to the archaic *H. sapiens* group. The thickness of the Xujiayao parietal fragments is within the range of *H. erectus* (Jia et al., 1979; Wu, 1980; Pope, 1992; Wu and Poirier, 1995). However, the mastoid angle for at least two of the parietals (nos. 6 and 10) are thinner than *H. erectus*, but thicker than modern humans. The mandibular fossa is as deep as *H. erectus*, but the glenoid process is more similar to Dali. The occipitals display a reduced torus region. A partial maxilla of a juvenile was also excavated from Xujiayao. Presence of a deciduous second molar in the maxilla suggests the individual died between 7 and 9 years of age (Jia et al. (1979). Permanent maxillary teeth measurements all fall within the range of *H. erectus*. Pope (1992, p 269) notes that overall, the Xujiayao materials "present a combination of characters of less developed ectocranial superstructures (a reduced occipital torus, reduced occipital angulation, reduced parietal curvature, and expanded cerebellar fossae) in conjunction with greater general robusticity in some specimens (thick cranial bones and large dentition) and probably more marked sexual dimorphism than is generally seen in modern populations." The importance of the Xujiayao material is that we are in a better position to understand the degree of morphological variation within a population from a single spatiotemporal point.

Homo cf. *erectus* and archaic *H. sapiens* fossils have been reported from at least three cave localities in North Korea: Yokpo Daehyundong, Dokchon Soongnisan, and Ryonggok (Park, 1992; Norton, 2000). Interestingly, North Korean paleoanthropologists have assigned these fossils to *H. sapiens neandertalensis* (e.g., Archaeology Research Laboratory, 1978; Kim et al., 1985; Jun et al., 1986). My own reading of the admittedly sparse North Korean literature indicates that North Korean paleoanthropologists refer all hominin fossils not clearly *H. erectus* or modern *H. sapiens* to *H. sapiens neandertalensis*. Two primary problems with accurately assessing the North Korean materials are the current access constraints and uncertainties about the chronometric ages (Norton, 2000). Currently, all non-North Korean scholars are restricted to the published literature and secondary or tertiary casts of the North Korean hominin fossils. The North Korean fossils date to the Middle to Late Pleistocene based usually on biostratigraphy, but occasionally a U-series or TL date has been reported (Norton, 2000). I present as much dental metric data as available from the Korean literature for all reported Pleistocene hominin fossils, including generally accepted modern

TABLE 2. Published raw data for Korean hominin fossil teeth[a]

Age	Site	Hominin affiliation	Specimen	Mesio-distal	Buccal-lingual
Middle/Late Pleistocene	Dokchon Soongnisan	Archaic *H. sapiens*	Lower right M1	11.6	10.5
Middle/Late Pleistocene	Dokchon Soongnisan	Archaic *H. sapiens*	Upper left M2	9.4	11.7
Late Pleistocene	Ryonggok	Modern *H. sapiens*	Mandible no. 1 left M1	10.5	11.8
Late Pleistocene	Ryonggok	Modern *H. sapiens*	Mandible no. 1 left M2	11.2	11.5
Late Pleistocene	Ryonggok	Modern *H. sapiens*	Mandible no. 1 left M3	9.8	10.6
Late Pleistocene	Ryonggok	Modern *H. sapiens*	Mandible no. 1 right M1	11.8	12.2
Late Pleistocene	Ryonggok	Modern *H. sapiens*	Mandible no. 1 right M2	11.7	12.2
Late Pleistocene	Ryonggok	Modern *H. sapiens*	Mandible no. 1 right M3	9.8	10.4
Late Pleistocene	Ryonggok	Modern *H. sapiens*	Mandible no. 2 left P3	6.8	8.7
Late Pleistocene	Ryonggok	Modern *H. sapiens*	Mandible no. 2 left P4	7.0	9.2
Late Pleistocene	Ryonggok	Modern *H. sapiens*	Mandible no. 2 left M1	12.2	11.6
Late Pleistocene	Ryonggok	Modern *H. sapiens*	Mandible no. 2 left M2	11.2	11.8
Late Pleistocene	Ryonggok	Modern *H. sapiens*	Mandible no. 2 left M3	11.8	11.4
Late Pleistocene	Ryonggok	Modern *H. sapiens*	Mandible no. 2 right M1	11.2	11.6
Late Pleistocene	Ryonggok	Modern *H. sapiens*	Mandible no. 2 right M2	12.6	11.8
Late Pleistocene	Ryonggok	Modern *H. sapiens*	Mandible no. 2 right M3	11.4	11.6
Late Pleistocene	Ryonggok	Modern *H. sapiens*	Mandible no. 6 M1	11.2	12.2
Late Pleistocene	Ryonggok	Modern *H. sapiens*	Mandible no. 6 M2	11.4	11.8
Late Pleistocene	Ryonggok	Modern *H. sapiens*	Mandible no. 6 M3	12.1	12.2
Late Pleistocene	Ryonggok	Modern *H. sapiens*	Skull no. 3 Upper left M1	10.4	10.5
Late Pleistocene	Ryonggok	Modern *H. sapiens*	Skull no. 3 Upper left M2	9.2	11.2
Late Pleistocene	Ryonggok	Modern *H. sapiens*	Skull no. 3 Upper left M3	8.2	10.2
Late Pleistocene	Ryonggok	Modern *H. sapiens*	Skull no. 3 Upper right M1	10.2	11.0
Late Pleistocene	Ryonggok	Modern *H. sapiens*	Skull no. 3 Upper right M2	9.8	11.2
Late Pleistocene	Ryonggok	Modern *H. sapiens*	Skull no. 3 Upper right M3	8.2	11.2
~30 ka	Kumchon	Modern *H. sapiens*	Mandible canine	7.2	8.0
~30 ka	Kumchon	Modern *H. sapiens*	Mandible P3	7.3	8.3
~30 ka	Kumchon	Modern *H. sapiens*	Mandible P4	7.2	8.2
~30 ka	Kumchon	Modern *H. sapiens*	Mandible M1	12.9	11.8
~30 ka	Kumchon	Modern *H. sapiens*	Mandible M2	12.3	11.7
Late Pleistocene	Mandalli	Modern *H. sapiens*	Lower left M2	12.4	11.4
Late Pleistocene	Mandalli	Modern *H. sapiens*	Lower right M2	12.3	11.4

[a] Once we ascertain whether or not the measurements and landmarks used by North Korean paleoanthropologists match Western terminology, we will be in a better position to present and analyze the cranial and postcranial data.

H. sapiens (Table 2). Although other metric data is available for the North Korean hominin fossils, how the measurements match western terminology is still being ascertained. I describe the Yokpo Daehyundong, Dokchon Soongnisan, and Ryonggok fossils below (see also Norton, 2000).

The Yokpo Daehyundong hominin fossils consist of frontal, occipital, and parietal fragments of a juvenile estimated to be 7–8 years of age at time of death (Fig. 2; Kim et al., 1985). However, because of a well-developed occipital torus and a pronounced supraorbital torus, characteristics similar to ZKD Locality 1 *H. erectus*, it may be possible the Yokpo hominin might be better allocated to the *H. erectus* taxon (Norton, 2000). If this holds up to further scientific scrutiny, Yokpo Daehyundong would be the most eastern locality yielding *H. erectus* yet known. The Dokchon Soongnisan hominin fossils were excavated from two different stratigraphic levels. In the lower stratigraphic level, two teeth (lower M1 and upper M2) and a scapula fragment were recovered, whereas in the upper layer a partial mandible retaining a P4 and M1 were excavated (Fig. 3). Bronze Age artifacts were present 50 cm above the stratigraphic level containing the partial mandible, (Archaeology Research Laboratory, 1978; Norton, 2000). The partial mandible displays clear evidence of a mental eminence, indicating clear affiliation with modern *H. sapiens*. Based on biostratigraphy, Yokpo Daehyundong and Dokchon Soongnisan were both dated to the Middle-Late Pleistocene (Norton, 2000).

Perhaps the most important Pleistocene hominin fossil site in North Korea is Ryonggok Cave, located near the present day capital city, Pyongyang (Jun et al., 1986; Norton, 2000). An assortment of hominin fossils were recovered (see Norton, 2000: his Table 3) representing at least five individuals from at least four separate stratigraphic levels. Jun et al. (1986) assigned Ryonggok to archaic *H. sapiens*, primarily based on the initial chronometric dating analysis. The chronometric dates are very disparate: initial TL = ~500–400 ka; later U-series = ~48–46 ka. Based on biostratigraphy, the associated archaeology, and the morphology of the hominin fossils, I suggest a Late Pleistocene age is a more reasonable one for the Ryonggok hominins (see also Chung, 1996; Norton, 2000). In particular, it should be noted that the Ryonggok hominin morphology includes "a rounded cranial vault, weak supraorbital tori, short face, steeply inclined forehead, absence of an occipital torus..., and presence of a chin" (Norton, 2000, p 814). Additionally, Skull no. 3 has an estimated cranial capacity of 1,650 cm^3, and Skull no. 7 is estimated to be 1,450 cm^3. Both are well within the range of modern humans and demonstrably outside the range of archaic *H. sapiens* (Fig. 4). Metric analysis of the mesiodistal and buccal-lingual measurements of the Ryonggok upper molars also generally fall within the range of modern humans (Fig. 5). Thus, it seems fairly clear that the Ryonggok hominins represent modern humans rather than archaic *H. sapiens*.

Interestingly, most of the Ryonggok hominin fossils appear to represent old individuals. For example, Mandible no. 1 is from a ~50-year-old female, Mandible no. 2 is thought to be from a ~45-year-old male, and Mandible no. 4 represents a ~40-year-old female (Jun et al., 1986). Although determining age of individual clearly past the



Fig. 2. Yokpo Daehyundong hominin fossils. [Color figure can be viewed in the online issue, which is available at wileyonlinelibrary.com.]

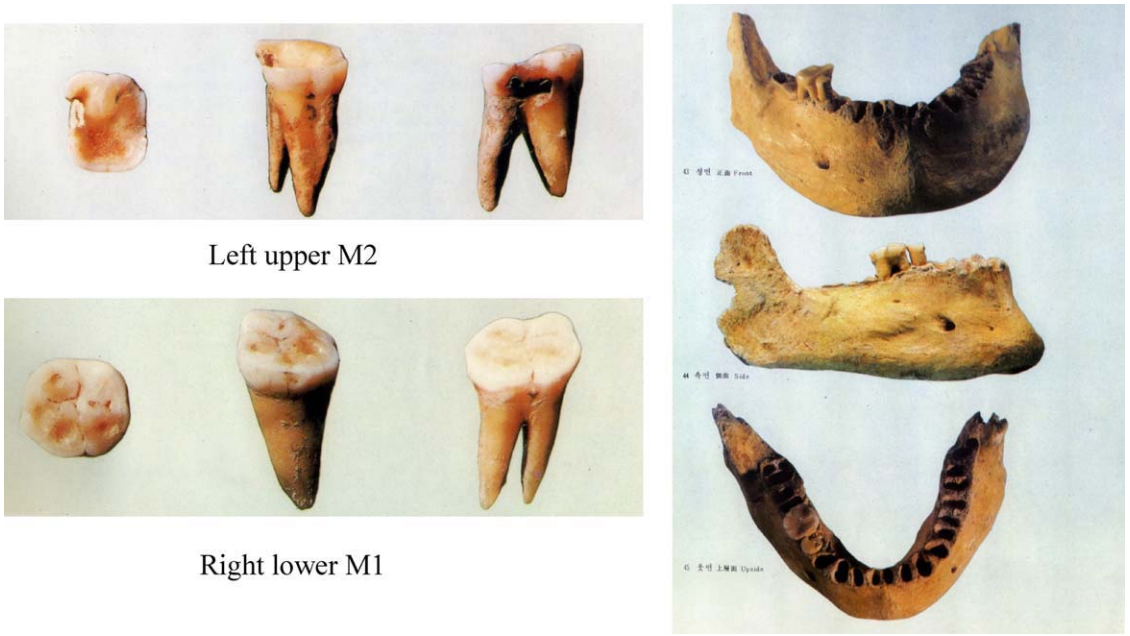


Fig. 3. Dokchon Soongnisan hominin fossils. Note the presence of a distinct mental eminence on the mandible. [Color figure can be viewed in the online issue, which is available at wileyonlinelibrary.com.]

juvenile stage to exact year should be treated cautiously, the human mandibles clearly represent fully mature or aged individuals. As such, the possibility that Ryonggok Cave may have been a Late Pleistocene burial site should at least be considered. Intriguingly, a bone artifact with perforations reminiscent of a human face was found in stratigraphic level 11, which is also the most abundant human fossil layer (Jun et al., 1986; Norton, 2000). More detailed studies of the site and materials are clearly needed. In particular, comparative study of the fossils and archaeology from Zhoukoudian Upper Cave and Xiaogushan in China (Norton and Gao,

TABLE 3. Seventeen cranial morphological traits compared and contrasted from archaic Homo sapiens fossils from China and Africa and 10 characters distinguishing China and Europe (after Wu, 1988b; Wu and Brauer, 1993)

China–Africa	China–Europe
1. Midsagittal elevation on the frontal squama	1. Midsagittal elevation
2. Bregmatic eminence	2. Flatness of the nasal saddle
3. Superior view of glabella region	3. Orientation of the anterolateral surface of the frontal process of the zygomatic
4. Midsagittal contour of the frontal squama	4. Contour of the lower border of the zygomatic process of the maxilla
5. Position of minimum distance between the superior temporal lines	5. Lower upper face
6. Interparietal sulcus	6. Shovel-shaped incisor
7. Angular torus	7. Position of the maximum breadth of the skull
8. Location of the maximum cranial breadth	8. Orbital margins
9. Lambdoidal ossicle	9. Sutures between the frontal and the nasal and maxillary bones
10. Protruding of the occipital region	10. Lambdoidal ossicle
11. Occipital torus	
12. The posterior part of the nuchal plane in front of the occipital torus	
13. The orientation of the anterolateral surface of the frontal process of the zygomatic bone	
14. Bulging lateral to the upper part of pyriform aperture	
15. The lower border of the zygomatic process of the maxillary bone	
16. Alveolar prognathism	
17. Upper facial height	

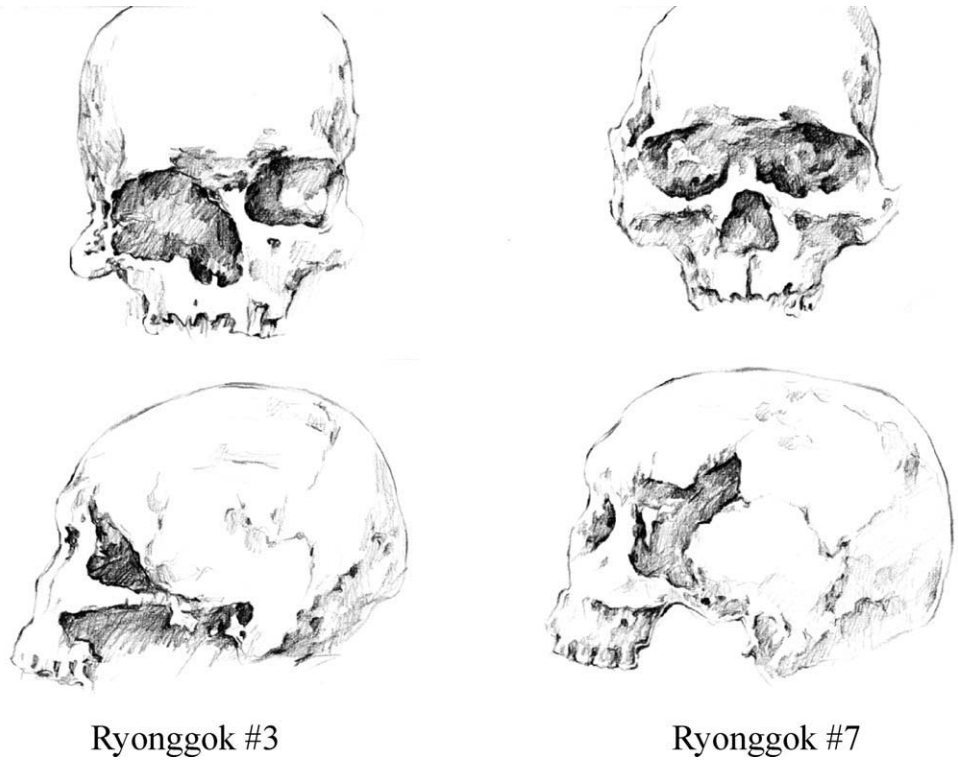


Fig. 4. Ryonggok hominin fossils. Fairly intact crania of modern *Homo sapiens*. Note the rounded vault, reduced supraorbital tori, and flattened occipital tori.

2008b; Norton and Jin, 2009) and Ryonggok Cave are warranted.

In 2006, a partial hominin calvarium was excavated from a pit 6-m deep during gold mining at the Salkhit site in northeastern Mongolia (Coppens et al., 2008). The hominin fossils are represented by a complete frontal bone and two partial parietal fragments. Although no chronometric dates are yet available, the associated fauna (e.g., *Coelodonta antiquitatis*) suggests a Late Pleistocene age (Coppens et al., 2008). Morphological analysis of the hominin cranial fragments revealed the presence of a mosaic of modern and archaic traits. Plesiomorphic characters include a frontal keel and relatively developed brow ridges, though clearly not as

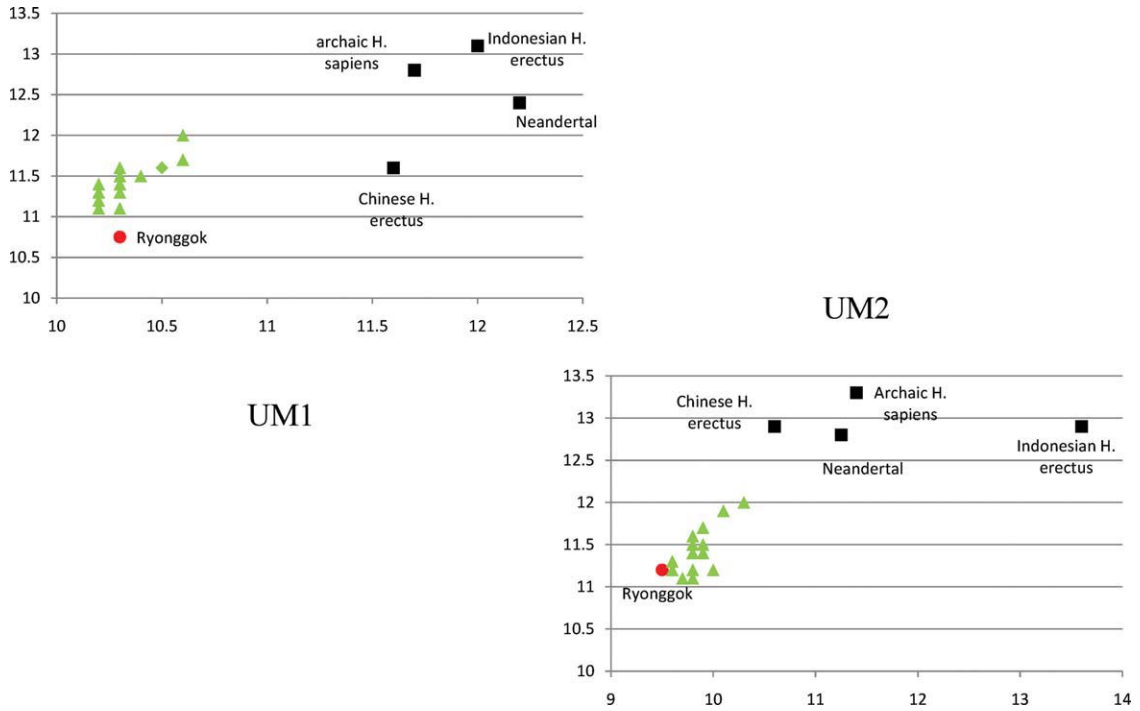


Fig. 5. Mesial-distal/buccal-lingual measurements for upper M1 (UM1) and upper M2 (UM2) Ryonggok human fossils (see Table 2 for raw data) compared with Holocene Chinese and different hominin taxa [MD along X axis; BL along Y axis; data for Holocene Chinese from Brace et al. (1984); data for various hominin taxa from Bailey and Liu (2010); all data are population averages]. [Color figure can be viewed in the online issue, which is available at wileyonlinelibrary.com.]

pronounced as *H. erectus*. Apomorphies include the absence of a sagittal keel on the parietals. According to Coppens et al. (2008), comparative metric analysis more closely aligns the Salkhit hominin with Neandertals, *H. erectus* sensu stricto, and archaic *H. sapiens* and away from modern humans of East Asian affinity. A number of Neandertal traits were identified, including the absence of the sphenoparietal sinus and superciliary arches that are thick at the frontal squama area and that taper off to the sides (Coppens et al., 2008). Given the recent argument that Neandertals have been identified as far east as Okladnikov Cave in southern Siberia (Krause et al., 2007), assigning the Salkhit hominin to Neandertals cannot be entirely discounted. This would be particularly interesting because recent morphometric studies of some modern human fossils from China (e.g., Tianyuandong and ZKD Upper Cave) suggest certain affinities with western Asian and European modern humans (Kamminga and Wright, 1988; Kamminga, 1992; Cunningham and Wescott, 2002; Cunningham and Jantz, 2003; Shang et al., 2007; Harvati, 2009). Although it has recently been suggested that evidence of modern human behavior moved west to east (Norton and Jin, 2009), it should be remembered that morphological and behavioral evolution do not always go hand in hand.

Important SE Asian hominin fossil localities

A number of important Middle Pleistocene *H. erectus* localities are present in China, including Tangshan Huludong, Hexian, Chenjiawo, and Yunxian (Etler and Li, 1994; Wu and Poirier, 1995). Other SE Asian *H. erectus* fossils include Tham Kuyen (Vietnam), Tam Hang (Laos), Had Pu Dai (Thailand), and Ngandong and Sambungmacan (Indonesia) (Olsen and Ciochon, 1990 Ciochon et al., 1996; Schepartz et al., 2000; Anton, 2003; Marwick, 2009), though the phylogeny of the hominin fossils from mainland SE Asia is still debated. The primary Middle Pleistocene archaic *H. sapiens* localities are Chaoxian (China), Maba (China), and Ma U'Oi (Vietnam). Besides Chaoxian and Maba, a number of archaic *H. sapiens* localities from southern China are described in Wu and Poirier (1995), including Changyang and Tongzi. Thum Wiman Nakin in Thailand is another potentially important locality that may eventually be considered an archaic *H. sapiens* site (Tougard et al., 1998). Reviews of the Paleolithic record of SE Asia published over the past two decades (e.g., Olsen and Ciochon, 1990; Pope and Keates, 1994; Schepartz et al., 2000; Marwick, 2009) indicate the great potential and need for more detailed multidisciplinary field and laboratory research programs in the region.

Chaoxian is an archaic *H. sapiens* locality situated in Anhui Province, eastern China and excavated in 1982 and 1983 (Wu and Poirier, 1995; Bailey and Liu, 2010). The Chaoxian hominin fossils are represented by a partial occipital, a partial maxilla, and three isolated maxillary teeth. Chaoxian is actually comprised of two distinct localities (A and B), separated by a 2–4 m limestone ridge. Locality A is represented by a typical Early/Middle Pleistocene fauna (e.g., *Hyaena brevirostris licenti* and *Meganterion* sp.), but no hominin fossils. Locality B has Middle Pleistocene fauna (e.g., *Hyaena brevirostris sinensis*, *Stegodon*, and *Sus xiaozhu*) and the archaic *H. sapiens* fossils (Shen et al., 2010). The initial U-series dates on associated animal teeth suggested an age range of 200–160 ka (Chen et al., 1987). However, a more recent study using TIMS U-series on associated speleothems indicated the hominin fossils could be bracketed between 360 and 310 ka or even older (Shen et al., 2010). Although the association between the dated

speleothem section and the hominin fossils cannot be made with full degree of confidence, minimally, the Chaoxian hominin fossils should date to the Middle Pleistocene. No hominin trace fossils (manuports, lithics, or hominin-modified bones) were recovered from the site during either the initial surveys or the ensuing excavations (Wu and Poirier, 1995).

The Chaoxian occipital and maxilla are described in detail by Wu and Poirier (1995) and the teeth by Bailey and Liu (2010). The occipital fragment includes a large section of the occipital squama. Although an estimate, the occipital angle falls between ZKD *H. erectus* and modern humans. The thickness of the center of the occipital torus (7 mm) falls below the range of ZKD *H. erectus*, except ZKD no. VIII (Wu and Poirier, 1995). Based on tooth eruption and wear, the estimated age of the individual is about 30 years (Wu and Poirier, 1995) or more conservatively, a young- to middle-aged adult (Bailey and Liu, 2010). Morphometric analysis indicates that the Chaoxian teeth are "unremarkable" in that they fall within range of other Middle Pleistocene hominins (Bailey and Liu, 2010). In general, the teeth are considered to be large, with an occlusal morphology retaining primitive features (e.g., developed P^4 accessory ridges, accessory fissures, and crests on the molars) of other Middle Pleistocene archaic hominins (Bailey and Liu, 2010). The anterior teeth are heavily worn and attributed to paramasticatory use, a trait found in many Neandertal fossils and other archaic *H. sapiens* (Brace, 1964; Brose and Wolpoff, 1971). The Chaoxian hominin teeth do not display any character traits that might align them with Neandertals (Bailey and Liu, 2010).

In 1958, local farmers excavated an archaic *H. sapiens* cranium from Shizishan near Maba village, Shaoguan Municipality, Guangdong Province, southeastern China (Wu and Poirier, 1995). Shizishan is represented by two low-lying limestone hills that has many cave entrances and a complicated network of naturally formed tunnels that stretch for several hundred meters at least. Separated by less than 100 m, it is thought that the two hills once formed a single hill. In the same cave that contained the hominin cranium, a partial hominin mandible and seven teeth were also discovered during fieldwork conducted in 1960 and 1984, respectively. The mandible and teeth are currently under study (Bae et al., n.d.). The age of the hominin fossils is unclear. The initial U-series analysis indicated an age between 135 and 129 ka (Yuan et al., 1986). However, a more recent TIMS U-series study of the same deposits suggested the capping flowstone should more readily date to 237 ka (Gao et al., 2007), thus pushing the minimum age of the Maba hominin back by at least 100,000 years. It should be noted, however, that the relationship between the capping flowstone and the hominin fossils is unclear; thus, caution is warranted when discussing the chronology of the Maba hominin (Bae et al., n.d.). Interestingly, *Palaeoloxodon namadicus*, a taxon usually found in higher latitudes was identified in the Maba faunal assemblage. No hominin trace fossils were found during subsequent survey or excavation (Wu and Poirier, 1995).

I restrict the following description to the Maba partial cranium, with more detailed description of the partial mandible and teeth presented later (Bae et al., n.d.). The Maba fossil is composed of a partial frontal, parietals, right orbit, and nasal region, thought to represent an adult male (Pope, 1992; Wu and Poirier, 1995). The Maba cranium is relatively flat, but higher than *H. erectus* (Wu and Wu, 1985; Wu and Poirier, 1995). The degree of postorbital constriction is pronounced. The arch-cord index of the parietal falls close to *H. erectus* and is much higher than in modern humans (Wu and Poirier, 1995). However, the Maba cranial walls are thinner than *H. erectus*. Perhaps the two most distinguishing characteristics of the cranium are the pronounced supraorbital tori, which are reminiscent of Neandertals and the rounded orbits, the latter feature which is not found in any other Early or Middle Pleistocene hominin from China (Wu and Peng, 1959; Wu and Wu, 1985). Although there have been suggestions that Maba could represent an eastern Asian Neandertal (e.g., Wu and Wu, 1985), there is general agreement that Maba is more similar to other archaic *H. sapiens* from China. Interestingly, the Maba partial cranium displays similarities with the Hathnora calotte from India, including rounded eye orbits, relatively robust supraorbital tori, and a flattened occipital region.

Ma U'Oi is a cave located in northern Vietnam just south of Hanoi, which was discovered during survey work at the nearby Chieng Xen cave site in 1999 (Demeter et al., 2004, 2005). Subsequent fieldwork at Ma U'Oi revealed the presence of two teeth (upper M2 and lower M1) and a partial occipital bone that were assigned to the archaic *H. sapiens* taxon. Ma U'Oi is situated in a tower karst region of the Annamitic Mountain Chain, with many additional caves located in the region (Demeter et al., 2005). The hominin fossils were found in the same stratigraphic level approximately 25–30 cm below the surface and in association with typical SE Asian *Ailuropoda-Stegodon* faunas (Bacon et al., 2004). Based on comparative analysis of the biostratigraphy from Thum Wiman Nakin (~169 ka)[4] and Lang Trang (~80–60 ka), Ma U'Oi is thought to date to the late Middle Pleistocene to Late Pleistocene (Demeter et al., 2005). U-series dating of associated speleothems indicated an age range between 193 and 49 ka, which largely substantiates the earlier biostratigraphic study (Bacon et al., 2006). No hominin trace fossils have been reported from the cave.

The Ma U'Oi teeth were assigned to archaic *H. sapiens* for the following reasons (Demeter et al., 2004, 2005). In general, heavily worn molars of *Pongo* and *Homo* can be difficult to distinguish (Ciochon et al., 1996). Although heavily worn, the Ma U'Oi lower M1 displays "asymmetrical distribution of the enamel thickness on the occlusal surface" (Demeter et al., 2004, p 538), a characteristic more indicative of *Homo* rather than *Pongo*. The mesiodistal and buccal-lingual measurements more readily fall within the range of *H. erectus* rather than *H. sapiens*. However, the lower M1 lacks taurodontism, a trait considered more characteristic of Neandertals and *H. erectus*. Primarily because of this mosaic of morphological characteristics and partly because of its assumed geological age, the Ma U'Oi lower M1 is considered to be from an archaic *H. sapiens* (Demeter et al., 2004). The upper M2 also displays an "irregular distribution of enamel thickness and the absence of a doubled crista transversa" (Demeter et al., 2005, p 397), thus the *Homo* rather than *Pongo* assignment. The upper M2 lacks any

[4]Based on a U-series date on an overlying capping speleothem with an age older than ~169 ka and the presence of *Crocuta crocuta ultima*, which replaced *Hyaena brevirostris sinensis* during the late Middle Pleistocene, the hominin fossil from Thum Wiman Nakin is considered to date to between 250 and 170 ka (Tougard et al., 1998).

evidence of taurodontism and occlusal wrinkles, suggestive of *H. sapiens* rather than *H. erectus* and *Pongo*. In addition, the mesiodistal and buccal-lingual measurements of the upper M2 fall within range of archaic *H. sapiens* (Demeter et al., 2005). The occipital fragment (upper left squamous region) is morphologically indistinguishable from archaic and modern *H. sapiens* (Demeter et al., 2005).

H. HEIDELBERGENSIS, ARCHAIC *H. SAPIENS*, OR SOMETHING ELSE

This review has raised a number of questions. Three of these are: 1) Should the eastern Asian hominin fossils be allocated to the *H. heidelbergensis* taxon? 2) Should the eastern Asian hominin fossils continue to be referred to as archaic *H. sapiens*? 3) Should another term be used to represent the eastern Asian Middle Pleistocene hominins that cannot be considered *H. erectus* (e.g., Gongwangling, Chenjiawo, ZKD Loc. 1, Yunxian, Hexian, and Ngandong) or modern *H. sapiens* (e.g., Tianyuandong, ZKD Upper Cave, and Ryonggok Cave)?

Should the eastern Asian hominin fossils be allocated to the *H. heidelbergensis* taxon?

It is becoming more widely accepted that most European and African Middle Pleistocene hominins can be assigned to *H. heidelbergensis*. However, in reviewing the case for allocating the eastern Asian Middle Pleistocene archaic *H. sapiens* into *H. heidelbergensis*, there do not seem to be many strong arguments. The primary cited reasons for allocating eastern Asian archaic *H. sapiens* into *H. heidelbergensis* can be grouped into two categories: 1) apomorphic traits originally assigned to eastern Asian archaic *H. sapiens* now appear in older western Old World *H. heidelbergensis* fossils, suggesting population movements from west to east; and 2) the possible overlap between *H. erectus* and archaic *H. sapiens* in eastern Asia suggests that one cannot be ancestral to the other. Perhaps the strongest cases for *H. heidelbergensis* are presented by Rightmire (1998, 2001, 2004) and Groves and Lahr (1994).

Rightmire's (1998) review of the evidence for *H. heidelbergensis* notes that specialists of Chinese paleoanthropology often use the presence of a canine fossa in the Chinese archaic *H. sapiens* fossils to distinguish them from western Old World hominin fossils. However, it seems that the older European Gran Dolina hominin fossils also display evidence of a canine fossa (Bermudez de Castro et al., 1997; Bischoff et al., 2007). It should be noted that Etler (1996) observed the presence of a canine fossa on the Yunxian *H. erectus* fossils, which may be penecontemporaneous with Gran Dolina or even older. In another study, Rightmire (2004) showed that *H. heidelbergensis* was significantly more encephalized than *H. erectus*. Interestingly, the Dali and Jinniushan crania displayed the same degree of encephalization as other members of the *H. heidelbergensis* hypodigm. I concur with Rightmire that the higher degree of encephalization in *H. heidelbergensis* vis-à-vis *H. erectus* is likely related to major behavioral innovations that seem to have occurred during the Middle Pleistocene. For example, Rightmire (2004) notes a number of cases of effective hominin hunting in Middle Pleistocene western Eurasia (e.g., Boxgrove, Schoningen). Our own taphonomic studies of the faunal remains from the archaic *H. sapiens* Xujiayao site in northern China (Norton and Gao, 2008a) support the argument that these hominins were efficient predators. However, increasing encephalization could more parsimoniously be explained as a case of parallel evolution.

Based on these arguments, Rightmire (2004, p 119) concludes that "the Asian hominins can be viewed as more closely related to other (western) Middle Pleistocene populations than to local, late-surviving *Homo erectus*" and that "[t]he spread of some populations of *Homo heidelbergensis* into the Far East cannot be ruled out" (Rightmire, 1998, p 225). In noting the possible chronological overlap of *H. erectus* and archaic *H. sapiens* populations in China, Groves and Lahr (1994) suggest that it is impossible that the latter group could be derived from the former. Indeed, Groves and Lahr (1994, p. 3) concluded that "*Homo erectus* at first existed by itself in China; *Homo heidelbergensis* then entered and the two coexisted for a time; finally *H. erectus* became extinct there, and *H. heidelbergensis* persisted alone: an early 'replacement' event."

Proponents of multiregionalism have long argued that some degree of gene flow existed more or less continuously between the different regions of the Old World (Wolpoff et al., 1984; Wu, 1988b; Pope, 1991, 1992; Wu and Brauer, 1993; Etler, 1996, 2004; Wolpoff, 1999; Hawks et al., 2000; Trinkaus, 2006). Assuming some degree of gene flow, it should come as no surprise then that a few morphological similarities are present between the eastern and western Old World "archaic" *H. sapiens* fossils, particularly where certain older character traits appear in one area of the world and then emerge later in other regions. A case in point may be the distinct canine fossa, which appears in the Gran Dolina *H. antecessor* fossils and may have later become a prominent feature of archaic *H. sapiens* in eastern Asia, though it should be noted that the character was identified in Yunxian *H. erectus* as well (Etler, 2004). Another well-known example is the shovel-shaped incisors of Neandertals, though the morphology of the shoveling appears to be different from the eastern Asian archaic *H. sapiens* (Crummett, 1994). On a fossil by fossil comparison, other morphological similarities do exist between the eastern and western Eurasian hominin fossils (Pope, 1992; Etler, 1996). For instance, the Dali and Jebel Irhoud 1 skull from Morocco display similarities, particularly in the "flat and broad mid-face, squared eye sockets, and keeled braincase" (Etler, 2004, p 46). However, Dali may be much older than originally proposed (GJ Shen, personal communication), thus suggesting a case of traits moving from east to west, rather than the reverse. Other alternatives are that the set of traits were present in the last common ancestor, that this is a case of parallel evolution, or that a case of *H. heidelbergensis* dispersing into eastern Asia (Groves and Lahr, 1994; Rightmire, 1998, 2004, 2008; Klein, 2009).

The occasional presence of a similar trait aside, however, a number of paleoanthropologists have argued that the Chinese archaic *H. sapiens* differs noticeably from penecontemporaneous hominins from Europe and Africa. The strongest cases have been presented by Wu (1988b; Wu and Brauer, 1993; Wu and Poirier, 1995), Pope (1991, 1992), and Etler (1996, 2004). These studies can be divided into two separate approaches: 1) presence/absence of distinct character traits; and 2) examination of morphological regions, particularly on the skull (e.g., midfacial region).

In a series of writings, Wu (1988; Wu and Brauer, 1993; Wu and Poirier, 1995) describes a set of 10 morphological cranial characters that distinguish Chinese archaic *H. sapiens* from European *H. heidelbergensis* and 17 traits that distinguish the Chinese hominins from African Middle Pleistocene hominins (Table 3). Characters that appear repeatedly as distinguishing the Chinese archaic *H. sapiens* from the African and European fossils are the orientation of the frontosphenoidal process of the zygomatic bone; upper facial height; maxillary shovel shaped incisors; Inca bones; and M3 agenesis.[5] Although Wu admits that certain characters do appear in the European and African specimens, he argues that they appear in lower frequency. For instance, Ndutu, Petralona, and Saccapastore *H. heidelbergensis* have Inca bones, but not only do these bones appear in higher frequency in the Chinese archaics (e.g., Dali, Dingcun, Xujiayao), they are shaped differently (Wu, 1988b; Wu and Poirier, 1995). Furthermore, the Chinese hominins (e.g., Dali) have lower facial height and lower upper facial index scores than Broken Hill and the European specimens, suggesting to Wu (1988b; Wu and Brauer, 1993) that Chinese archaic *H. sapiens* have relatively short faces. Even though some Neandertals have maxillary shovel-shaped incisors, Wu (1988b; Wu and Poirier, 1995) argues that all Chinese hominin fossils display this character. It is interesting to note that the Chinese archaic *H. sapiens* are characterized by the presence of a flat nasal saddle, which differs markedly from the Neandertals who are well known for having a protruding nose (Wu and Poirier, 1995). Analysis of the presence/absence of individual traits has limitations, particularly because it is clear that many traits are interrelated (see discussion below).

Other paleoanthropologists have relied more heavily on the combination of traits in particular anatomical regions and the frequency with which the combination of traits appear across time and space, rather than single morphological characters. For example, in his review of the evidence for modern human origins in China, Pope (1992, p 251) chose to "emphasize a few anatomical regions which seem to best display variations between the Premoderns and Chinese *Homo erectus*: i.e., the face (in frontal and lateral aspects), the glenoid region (in basal aspect), and the occipital region (in posterior and lateral aspect)." In particular, the midfacial region is considered to be useful for studying inter-regional population and evolutionary morphological variation related to behavioral changes, though some of the traits may be correlated with the biomechanics related to mastication (Brace, 1967; Gill et al., 1988; Brace and Hunt, 1990). The morphology of the midfacial region in Asian hominins differs from European and African Middle Pleistocene hominins (Pope, 1992). For instance, when viewed from norma frontalis, Asian midfacial regions (zygion–zygion/nasion–prosthion) are characterized by "much smaller upper and lower midfaces with more horizontally oriented zygomatic bones, pronounced and more medially situated malar tubercles, a distinct incisura malaris, a more acute and inferiorly situated zygomaticomaxillary angle and a vertically shorter maxilla" (Pope, 1991, p 189). Petralona, Arago, Atapuerca, Bodo, and Broken Hill do not display this Asian midfacial morphology (Pope, 1991, 1992).

Most paleoanthropologists, including advocates of the *H. heidelbergensis* taxon, note that the eastern Asian archaic *H. sapiens* are generally morphologically distinct from the western Old World hominin fossils that have been assigned to *H. heidelbergensis* (e.g., Pope, 1992; Wu and Brauer, 1993; Rightmire, 1998, 2008; Etler, 2004; Klein, 2009). For example, when examining the general shape of the face, the European and African *H. heidelbergensis* have much larger faces than the eastern Asian archaics. In addition to the midfacial region, which is distinct from most western Old World *H. heidelbergensis* (Pope, 1991, 1992), the teeth of these hominin fossils are distinctive as well. In this case, Bailey and Liu (2010) and Demeter et al. (2005) found that the Chaoxian and Ma U'Oi teeth could not be classified with the western Old World *H. heidelbergensis* and/or Neandertals. Cartmill and Smith (2009, p 336) concluded recently that "the term 'Heidelberg' can be used with little hesitation for the European, Near Eastern, and African specimens. Whether the term should be extended to samples and specimens further east in Asia is a matter of choice." Even strong advocates of *H. heidelbergensis* (e.g., Rightmire, 2008) have had difficulty finding the justification to allocate the eastern Asian material to this taxon.

Should the eastern Asian hominin fossils continue to be referred to as archaic *H. sapiens*?

"Rarely indeed, however, have paleontologists ever found it necessary to distinguish between 'archaic' and 'anatomically modern' types of the same species, and there seems scant justification for squeezing these distinct hominid morphs into a single species. In any group other than Hominidae the presence of several clearly recognizable morphs in the record of the middle to upper Pleistocene would suggest (indeed, demonstrate) the involvement of several species." (Tattersall, 1986, p 170)

Tattersall (1986, p 165) stated that "ranges of morphological variation in closely related species in the living fauna normally overlap substantially or completely in most characters; some closely related species cannot be distinguished on the basis of hard parts." This led Tattersall to argue that paleoanthropologists are seriously underestimating the number of hominin species present in the Middle and Late Pleistocene. Tattersall rails against lumping all Middle Pleistocene taxa not readily allocated to *H. erectus* into *H. sapiens*, when he writes "[t]he 'grade'... is one of the most destructive canards that paleoanthropology has ever seen fit to inflict upon itself: a meaningless and undefined concept, apparently leaning heavily on brain size, that can be used to entomb all kinds of morphological loose ends and thus eliminate the need to examine them" (Tattersall, 1986, p 173). The particular problem Tattersall (1986) sees is that closely related species (e.g., various *Homo* taxa) share many morphological character traits, which, in many cases, overlap in range and frequency. Although much has changed in paleoanthropology in the almost quarter-century since Tattersall's article (e.g., since ca. 1990, the number of distinct hominin taxa has almost doubled), it would seem that he maintains the same negative opinion of the designation "archaic *H. sapiens*." Indeed, Tattersall and Schwartz (2008, p 54) concluded recently that "it is evident that virtually all of those hominid fossils

[5]Although M3 agenesis is not listed by Wu, it is well-known among paleoanthropologists to be considered a defining character of the Chinese hominin fossils that is evidence of continuity from *H. erectus* to modern Chinese (Wolpoff et al., 1984; Pope, 1991; Etler, 2004).

whose exact historical significance has been obscured by their assignment to the all-embracing wastebasket of 'archaic *Homo sapiens*,' in fact belonged to an array of separate biological entities, none of them evidently closely affiliated to living *Homo sapiens*." In Tattersall's view, these different Middle Pleistocene taxa should be classified as distinct species.

Nevertheless, it might be argued that giving latinized specific names to different fossils subconsciously implies reproductive isolation. Assuming that the evolutionary development of *H. heidelbergensis* occurred sometime during the Middle Pleistocene, there is little to no evidence that it is a distinct biological species in the sense that it could not have interbred with the other penecontemporaneous *Homo* taxa. In other words, there is little justification to argue that *H. heidelbergensis* could not have interbred with *H. erectus*, *H. antecessor*, or eastern Asian archaic *H. sapiens*. Indeed, the recent analysis of Neandertal DNA, which indicates some degree of admixture with modern *H. sapiens* is a case in point (Green et al., 2010). I am not taking the extreme stance that all *Homo* taxa should be assigned to the *H. sapiens* hypodigm (e.g., Wolpoff et al., 1994; Curnoe and Thorne, 2003). Morphological variation does appear across time and space, and it could and should be classified (see also Rightmire, 1998, 2004, 2008). I concur with Cartmill and Smith (2009, p 335) when they write "there is nothing wrong in principle with defining *H. heidelbergensis* as an intermediate evolutionary grade, separated from an ancestral species defined by symplesiomorphies and one or more descendant species defined by synapomorphies."

If we view *H. heidelbergensis* as a separate species, based primarily on chronophenetic grounds (sensu Pope, 1992; Cartmill and Smith, 2009), then there is strong justification for classifying *H. heidelbergensis* as a distinct grouping (as a morphospecies) and appropriate for the western Eurasian and African late Middle Pleistocene hominins. However, it might be a useful exercise to consider these late Middle Pleistocene hominins allotaxa for *H. heidelbergensis* (western Eurasia and Africa) and archaic *H. sapiens* (eastern Asia) (see allotaxa discussion in Jolly, 2001). Anton (2003) made a similar argument for referring to the different *H. erectus* sensu lato groups as allotaxa. Indeed, Anton (2003, p 126) notes that "such a view allows us to focus on the adaptations and biology of local groups, including questions of biogeographic isolation and local adaptation," rather than spending too much time on "unresolvable species debates."

It is true that archaic non-hominin taxa have yet to be identified (e.g., no one to my knowledge has argued for archaic *Pan paniscus*). In other realms of paleontology, minor differences are often grounds for coining a new specific name (see Tattersall, 1986). If we follow this argument, and based on chronophenetic differences between eastern Asian archaics and western Old World Heidelbergs, we might consider giving the former population a new species name rather than continue to refer to them as archaic *H. sapiens*. I address this question further below.

Should another term be used to represent the eastern Asian Middle Pleistocene hominins that cannot be considered *H. erectus* or modern *H. sapiens*?

Whatever shall we call late Middle Pleistocene eastern Asian hominins? Archaic, early, or premodern *H. sapiens* are the terms used most frequently to refer to the eastern Asian hominins that cannot be allocated to either *H. erectus* or modern *H. sapiens*. If we take the splitter approach, then we might consider using the term *H. daliensis* to refer to the archaic *H. sapiens* in NE Asia and *H. mabaensis* to refer to penecontemporaneous hominins in SE Asia, simply based on chronophenetic and biostratigraphic grounds (see above discussion). Morphological studies of the Zhoukoudian and Indonesian *H. erectus* fossils found similar regional variation (e.g., Anton, 2003). However, in the latter case, there are no current arguments to split these eastern Asian erectine populations into distinct species. The same might be argued for the archaic *H. sapiens* from eastern Asia. There is currently little justification for splitting eastern Asian archaic *H. sapiens* into two separate species.

Nevertheless, I suggest adding more species names to the already long list of Middle Pleistocene hominin taxa defeats the purpose of trying to better understand the dynamics involved with human evolution in eastern Asia. Until the eastern Asian human evolutionary record is better known, I suggest that it is better to err on the side of caution. A more parsimonious approach may be to continue to refer to these hominins as archaic *H. sapiens*, and in terms of their regional variation, as Dali man and Maba man. This is not a new observation in hominin paleontology. More than one half century ago, the venerable Ernst Mayr (1950, p. 115) made the same suggestion when he wrote "[v]ernaculars, such as 'Steinham man' or 'Piltdown man,' are just as useful and much less misleading. The formal application of generic and specific names simulates a precision that often does not exist." Indeed, the title of the recent Yin et al.'s (2002) study of the Dali deposits includes the nonspecific phrase "Dali Man."

It will likely continue to be argued that allocating all of these transitional specimens to archaic *H. sapiens* might obscure specific variation (e.g., Tattersall, 1986; Tattersall and Schwartz, 2008). However, for much the same reasons as the European and African fossils are now being allocated to *H. heidelbergensis* (based specifically on gradistic changes), nomenclature like *H. daliensis* and *H. mabaensis* could be argued to represent the northeast and SE Asian late Middle Pleistocene hominin fossils. Alternatively, we could simply use *H. mabaensis* to represent all of the fossils traditionally allocated to eastern Asian archaic *H. sapiens* because it was one of the first archaic *H. sapiens* fossils found and well studied. Ultimately, *H. daliensis* and *H. mabaensis* represent the same exact hominin fossils currently allocated to eastern Asian archaic *H. sapiens*, which begs the question: Does the designation of specific names in this case really represent more "exact" precision? It should be noted that I am not suggesting we sink western Eurasian and African *H. heidelbergensis* back into archaic *H. sapiens*. I am arguing, however, that we clearly need to increase the sample size of the eastern Asian hominin fossils to more fully understand the nature of hominin morphological variation during this spatiotemporal point.

DISCUSSION

"[T]here are no unique defining characters for Heidelbergs-just various mosaics of features that are not the same for all regions, or even for all specimens within a region." (Cartmill and Smith, 2009, p 335).

No true autapomorphic characters specific to *H. heidelbergensis* that crosscut time and space have been

identified (Rightmire, 1998, 2008; Etler, 2004; Cartmill and Smith, 2009). Thus, it is perhaps not surprising that opinions vary on what *H. heidelbergensis* actually represents and whether it should be considered to be present in eastern Asia during the Middle Pleistocene. The crux of the problem is that different schools of thought emphasize different approaches, resulting in widely varying interpretations. In particular, cladistic studies and which chronometric dates are considered seem to influence the debate over whether *H. heidelbergensis* is present or absent in eastern Asia. I discuss each of these points in turn below.

Cladistic studies

Cladistics has frequently been used to address questions concerning hominin phylogenetics (e.g., Eldredge and Cracraft, 1980; Tattersall, 1986; Groves, 1989; Harrison, 1993; Kitching et al., 1998; Collard and Wood, 2000; Wood and Lonergan, 2008) and recently in archaeology (e.g., O'Brien and Lyman, 2003; Lycett, 2007). Cladistics was originally developed by Hennig (1966) in the 1940s for entomological research. Basically, cladistics can be defined as the "analysis of the characters of organisms to infer the evolutionary branching sequence of a group's phylogeny" (Ashlock, 1974, p. 81). Cladistics organizes "things (be they species, populations, or artifacts) into a hierarchical pattern that reflects closeness of relationship based on the attributes (e.g., genes or morphology) exhibited by the individuals within those groups" (Lycett, 2007, p. 543–544). For example, presence/absence of a variety of cranial morphological traits (e.g., mental eminence, canine fossa, flattened nasal saddle, pronounced supraorbital tori, occipital bun, etc.) is often included in cladistic analyses. Cladistics has proven effective at distinguishing family-level categories, primarily because the "last common ancestors of the major groups are sufficiently distantly removed in time to allow the recognition of major adaptive patterns that characterize the extant representative" (Harrison, 1993, p. 348). Nevertheless, cladistic approaches have proven to be less robust in distinguishing variation at the lower generic–specific level categories.

Although Harrison (1993) rightly cites the integrity of the fossil data as a major hindrance in hominin phylogenetic reconstructions using cladistics methods, other problems do exist. For example, one apparent weakness of the cladistic approach is the question of the independence of characters and the inability of cladistics to be able to distinguish independent traits (Pope, 1992; Harrison, 1993; Groves and Lahr, 1994; Lieberman, 2008; Tattersall and Schwartz, 2008). Cartmill and Smith (2009, p. 330) justifiably observe that "[i]n any anatomical comparison in which different features are being tallied and contrasted, there is a tendency to think of all the items in the tally as separate entities produced independently by evolutionary forces." Pope (1992, p. 245) notes that "[i]t is of the utmost importance to establish that traits are independent of each other in order to prevent the generation of long lists of separate traits which are really manifestations of the same selection pressures or developmental complexes." Interestingly, Lieberman and colleagues, in a series of studies (e.g., Lieberman, 1995, 1998, 2008; Lieberman et al., 2000, 2002), have shown that the overall morphology of the human skull will depend on functional requirements involved minimally with speech, respiration, locomotion, mastication, and cognition. Indeed, Lieberman (2008, p. 56) observes that "[s]kulls are complex, strongly integrated structures characterized by high levels of covariation among multiple structures, even in different regions such as the face, basicranium, and neurocranium." A well-known example of this is the long-term influence on cranial morphology of a diet that relies more heavily on meat and cooked foods (including meat) (Aiello and Wheeler, 1995; Wrangham et al., 1999). Meat and cooked foods are softer and easier to chew, thus leading to less pressure placed on the overall masticatory apparatus. Less pressure leads to smaller teeth and reduced masseter and temporalis muscles and muscle attachments, which ultimately leads to a more orthognathic face and reduction or disappearance of the sagittal crest (as reviewed recently by Lieberman, 2008 among others). This implies that it is difficult to determine a trait that can be considered independent of other characters, and at the same time independent of any range of functions the skull is involved at any given time (Lieberman et al., 2002).

The key to understanding the dependence/independence of physical traits is to determine their genetic correlates (Cheverud, 1988; Pope, 1992; Etler, 2004; Pearson, 2004). It might be added that ascertaining possible epigenetic correlates might be a useful future exercise as well. Fortunately, the recent studies by Roseman, Weaver, and colleagues (e.g., Roseman, 2004; Roseman and Weaver, 2004; Harvati and Weaver, 2006a,b; Weaver et al., 2007, 2008; Weaver and Roseman, 2008) that examine the relationship between cranial morphology and neutral genetic variation have begun to shed light on the subject (see also recent studies by Betti et al., 2009, 2010; Smith, 2009; von Cramon-Taubadel, 2009a,b). In general, Roseman, Weaver, and colleagues found that population history accounts for roughly 50% of cranial morphology in modern humans, though it should be noted a great deal of variation exists. However, in some cases, they found that certain morphological features could not be readily explained by neutral genetic expectations. For example, Roseman (2004) found a strong correlation between cold climate and a cranial vault shape that was brachycephalic, as well as certain aspects of the nasal morphology. This finding led Roseman (2004) to conclude that the cranial morphology of the Siberian population in his data set was the result of natural selection; in this case, adaptation to extreme cold climates (see also Relethford, 2004). Furthermore, Ackermann and Cheverud (2004) found evidence that natural selection played a role in shaping facial variation between australopiths and early *Homo*. Interestingly however, Weaver et al. (2007) observed that rather than natural selection, the simple long-term effect of genetic drift could more readily explain the morphological variation seen in Neandertals and modern humans. Future studies may also shed light on the influence of natural selection and/or genetic drift on the morphological variation of the eastern Asian hominin fossils. Irrespective of whether natural selection or genetic drift is more influential, results of these studies suggest that genetic correlates need to somehow be factored into any cladistic analysis of human morphological variation. Ultimately however, despite the strengths of cladistic analyses, it is important to keep in mind that identifying species groupings should be seen "as an initial step in a cladistic analysis," rather than "as a method to identify species groupings" (Harrison, 1993, p 362).

Dating issues

In the early 1990s, it was proposed that *H. erectus* and *H. sapiens* overlapped in China (Chen and Zhang, 1991;

Chen et al., 1994; Groves and Lahr, 1994).[6] The primary evidence for this was the initial dating of the Hexian *H. erectus* fossils at ~190–150 ka and the nearby Chaoxian archaic *H. sapiens* fossils, which were initially dated to ~200–160 ka (Chen and Zhang, 1991). As an extension of these studies, some Chinese scientists (e.g., Chen et al., 1994) interpreted this supposed chronological overlap as support for the multiregional continuity model of modern human origins. As mentioned above, Groves and Lahr (1994) interpreted the chronological overlap to represent evidence of an early "replacement" event where invading *H. heidelbergensis* replaced the indigenous *H. erectus* in China. Thus, in Groves and Lahr's (1994) view, there is no evidence of regional continuity.

More recent chronometric dating analyses of many of the archaic *H. sapiens* localities in China indicate that the picture is much more complicated. A good example of this is the redating of the Hexian and Chaoxian sites. Grun et al. (1998) redated the Hexian *H. erectus* site using a combination of electron spin resonance and U-series techniques and derived an average age of 412 ka. Shen et al. (2010) recently redated the Chaoxian archaic *H. sapiens* site using the TIMS U-series method and calculated an age bracket that places the hominin fossil between 360 and 310 ka. An average age for the Jinniushan archaic *H. sapiens* locality has also been pushed back to ~260 ka (Chen et al., 1994; Lu, 2003; Rosenberg et al., 2006) and Maba back to ~237 ka (Gao et al., 2007). These more recent chronometric studies led Shen et al. (2010, p 27) to conclude that these new dates indicate "a *H. erectus*-archaic *H. sapiens* interface in China much earlier than 200 ka." In Shen et al.'s (2010) view, this *H. erectus*-archaic *H. sapiens* interface can now be pushed back to ~400–350 ka, particularly with the recent redating of the Zhoukoudian Locality 1 *H. erectus* site suggesting a minimum age of ~400 ka (Shen et al., 2009). These more recent redating analyses do not clarify the question of whether *H. heidelbergensis* dispersed into eastern Eurasia. However, the new dating studies do suggest that the evolution of archaic *H. sapiens* (or the movement of *H. heidelbergensis* into the region) is more consistent with the nature of the Middle Pleistocene human evolutionary record in the western Old World, where *H. erectus* begins to disappear and *H. heidelbergensis* appears. However, as discussed above, questions do exist about the relationship between the samples used for dating and the hominin fossils from these sites (see also Pope, 1992). Furthermore, the chronometric ages of Xujiayao, Dali, Dokchon Soongnisan, and Salkhit are not clear.

With constant improvements in the quality of chronometric dating studies, the chronological framework in which we place the hominin fossil record is constantly being adjusted. For now, however, it would probably be better to err on the side of caution when trying to interpret the eastern Asian hominin fossil record based primarily on supposed chronological similarities and/or differences. Although it falls outside the scope of this article, a good example of this is the dating debate of the Yuanmou *H. erectus* teeth. For example, it has been suggested recently that the Yuanmou hominin fossils date to ~1.7 Ma (Zhu et al., 2008), though others (e.g., Hyodo et al., 2002) had earlier suggested it date to the Middle Pleistocene. However, a number of researchers (e.g., Pope, 1988; Schepartz et al., 2000; Dennell, 2009) have questioned interpretations of the Yuanmou dates based solely on questions about context (i.e., some suggest the material was actually surface collected). Although making a general point, Klein (2009:XX) probably puts it most succinctly when he notes "that a first occurrence should be treated as a possible accident and even a second should be regarded as a possible coincidence. Only repeated, independent, mutually consistent occurrences can document a reliable pattern." If the book is to be firmly closed on Yuanmou, then similar vertebrate and trace fossils from similar spatiotemporal facies need to be found. The same can be said about many of the eastern Asian Middle Pleistocene sites mentioned here. A combination of consensually agreed on dating methods (both absolute and relative) need to be applied in concert on sites and materials from stratified deposits to be able to more confidently reconstruct the human evolutionary record in the region.

CONCLUSIONS

It is generally accepted in paleoanthropology today that advocating the extreme version of either the Replacement Hypothesis or Multiregional Hypothesis to explain the origin of modern humans throughout the Old World carries little weight (Lieberman, 2008; Pearson, 2008 among others). Rather, the degree of admixture between dispersing modern humans with indigenous hominins in places like Europe and eastern Asia is receiving greater attention in paleoanthropology. Indeed, even once hard line multiregionalists (e.g., Wu and Poirier, 1995) have moved away from the edge and now support some degree of admixture between modern humans from Africa and indigenous archaic *H. sapiens* in East Asia (Wu, 2004). The question of whether eastern Asian archaic *H. sapiens* should be classified as *H. heidelbergensis* can also be viewed in the light of dispersing hominin populations. In particular, if *H. heidelbergensis* dispersed from the western Old World and into eastern Asia some time during the Middle Pleistocene, then it would support the hypothesis that a third major dispersal event out of Africa occurred, as postulated by Templeton (2002, 2005). The first major dispersal would be classified as Out of Africa I, when *H. erectus* sensu lato moved out of Africa sometime between 2 and 1 million years ago. In Templeton's scenario, the second major dispersal event occurred sometime around 650 ka and involved some type of advanced *H. erectus* or archaic *H. sapiens*. In this case, western Old World archaics are allocated to *H. heidelbergensis*. The third event would be the dispersal of modern humans some time between 200,000 and 100,000 years ago. Thus, it should not be too surprising that certain character traits appear initially in the western Old World and later in eastern Asia. Traits that appear initially in eastern Asia and then in the western Old World could also be interpreted as evidence of gene flow in the opposite direction (Pope, 1992), common ancestry or parallel evolution.

No clear autapomorphic traits exist in the *H. heidelbergensis* hypodigm that clearly distinguish it from *H. erectus* sensu lato and *H. sapiens*. Furthermore, no clear similarities exist between western Old World *H. heidelbergensis* and archaic *H. sapiens* from eastern Asia, except perhaps some individual morphological character

[6] In their reanalysis of the Jinniushan age, Chen et al. (1994) interpreted the Jinniushan hominin as "modern" *H. sapiens*, rather than "archaic," although most paleoanthropologists support the latter classification (e.g., Wu and Poirier, 1995; Rosenberg et al., 2006).

traits on a fossil by fossil basis. In some cases, chronometric ages overlap, though these often vary depending on which dating technique is used. Thus, rather than allocating all eastern Asian Middle Pleistocene archaic *H. sapiens* to *H. heidelbergensis* or going to the extreme and assigning them new species names (e.g., *H. daliensis*, *H. mabaensis*), I suggest the eastern Asian hominin fossils should continue to be referred to as archaic *H. sapiens*. It will likely continue to be argued by some that archaic *H. sapiens* is a "wastebasket" category (e.g., Tattersall, 1986; Tattersall and Schwartz, 2008). However, it could be argued that the lack of clear autapomorphic characters that clearly define *H. heidelbergensis* is also grounds for considering it a wastebasket category as well.

I am not taking a side here on whether eastern Asian archaic *H. sapiens* should be viewed as strictly transitional between Asian *H. erectus* and modern *H. sapiens*. However, even small amounts of gene flow from dispersing *H. heidelbergensis* groups into eastern Asia during the Middle Pleistocene is probably the most parsimonious explanation as to why similar morphological features occasionally appear among penecontemporaneous western and eastern Old World hominins. I also suggest that eastern Asian hominin population size during the Middle Pleistocene was likely lower than in many regions of the western Old World, particularly Africa (sensu Wolpoff et al., 1984; Lycett and Norton, 2010). Although I am not taking the hard line stance that Africa and Eurasia were isolated from each other throughout the Middle Pleistocene (e.g., Dennell, 2009), I do suggest that smaller initial population sizes and the long-term effect of genetic drift most likely explain the morphologies of the Middle Pleistocene hominins from eastern Asia.

Ultimately, with the increase in multidisciplinary field and laboratory paleoanthropological research projects in eastern Asia (Norton and Jin, 2009, 2010), scientists will eventually have greater access to a much more extensive hominin fossil data set. By applying a combination of absolute and relative dating techniques to many of these important hominin fossil localities and linking the archaeological and paleoenvironmental records, we will be in a much better position to reconstruct the nature of human evolution in eastern Asia during the Quaternary. As the quality and quantity of the eastern Asian paleoanthropological data improve and increase, we will be in a much stronger position to address many of the questions that were raised in this review. This not only applies to any potential contribution the eastern Asian data have to the *H. heidelbergensis*/archaic *H. sapiens* question, but other paleoanthropological debates that cross-cut time and space. The scientists intrepid enough to take on this bold task will be the ones to figure out the true meaning of the eastern Asian paleoanthropological record.

The two primary conclusions drawn from this review are as follows:

1. Little evidence currently exists in the eastern Asian Middle Pleistocene hominin fossil record to support their assignment to *H. heidelbergensis*. Not allocating all of the eastern Asian hominins to *H. heidelbergensis* avoids false precision at a broad level (lumping).
2. Rather than add to the growing list of hominin fossil taxa by employing taxonomic names like *H. daliensis* for Northeast Asian fossils and *H. mabaensis* for SE Asian fossils, it is better to err on the side of caution and continue to use the term archaic *H. sapiens* to represent all of these hominin fossils. Avoiding giving formal Latin names to the different hominins that were present in eastern Asia during the Middle Pleistocene avoids false precision at the other end of the spectrum (splitting).

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